

BMATS101 – MATHEMATICS-I (CSE)

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MODULE 2

SERIES EXPANSION & MULTIVARIABLE CALCULUS

SERIES EXPANSION (FOUNDATION + APPLICATIONS)

1. Introduction to Series Expansion

In computer science and engineering, many functions are **too complex to compute directly**. Hence, they are approximated using **infinite series**.

Series expansion is widely used in:

- Numerical computation
- Error analysis
- Computer programming
- Approximation algorithms

This module introduces **Taylor's and Maclaurin's series** and their applications.

2. Functions and Expansions

A function $f(x)$ can be expressed as an **infinite sum of algebraic terms** about a point.

General idea:

$$f(x) \approx \text{Polynomial in } x$$

The two most important expansions are:

- **Taylor series**
- **Maclaurin series**

3. Taylor's Series

3.1 Statement of Taylor's Theorem

If a function $f(x)$ and its derivatives are continuous in the neighborhood of $x = a$, then:

$$f(x) = f(a) + (x - a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \dots$$

3.2 Important Points

- Expansion is about the point $x = a$
- Accuracy increases as higher-order terms are included
- Used to approximate values near $x = a$

4. Maclaurin's Series

4.1 Definition

Maclaurin's series is a **special case of Taylor's series** about $x = 0$.

4.2 Formula

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \dots$$

5. Standard Maclaurin Series (VERY IMPORTANT)

These must be **memorized for VTU exams**.

5.1 Exponential Function

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

5.2 Trigonometric Functions

$$\begin{aligned}\sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\end{aligned}$$

5.3 Logarithmic Function

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \quad (|x| < 1)$$

5.4 Binomial Expansion

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$$

6. Solved Problems (Maclaurin Series)

Example 1

Expand e^x using Maclaurin's series up to x^4 .

Solution

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}$$

Example 2

Expand $\sin x$ up to x^5 .

Solution

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

7. Expanding a Function Using Standard Series

Example 3

Expand $\log(1+x)$ up to x^3 .

Solution

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3}$$

8. Approximation Using Series

Series are used to find **approximate numerical values**.

Example 4

Find approximate value of $e^{0.1}$.

$$e^{0.1} \approx 1 + 0.1 + \frac{(0.1)^2}{2} = 1.105$$

9. Application in Computer Science

9.1 Programming

- Evaluating exponentials
- Computing trigonometric values
- Handling floating-point approximations

9.2 Error Approximation

- Truncation errors
- Rounding errors
- Precision control

INDETERMINATE FORMS & L'HOSPITAL'S RULE

1. Introduction to Indeterminate Forms

While evaluating limits, sometimes **direct substitution** gives expressions that do not lead to a definite value. Such expressions are called **indeterminate forms**.

Example:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{0}{0}$$

Here, the form $\frac{0}{0}$ is **indeterminate**.

2. Indeterminate Forms (VTU List)

The commonly occurring indeterminate forms are:

1. $\frac{0}{0}$
2. $\frac{\infty}{\infty}$
3. $0 \cdot \infty$
4. $\infty - \infty$
5. 0^0
6. ∞^0
7. 1^∞

3. L'Hospital's Rule

3.1 Statement of L'Hospital's Rule

If:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

results in an indeterminate form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, and if:

- $f(x)$ and $g(x)$ are differentiable near $x = a$
- $g'(x) \neq 0$

then:

$$\boxed{\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}}$$

If the result is still indeterminate, L'Hospital's rule can be applied **repeatedly**.

4. Solved Problems – $\frac{0}{0}$ Type

Example 1

Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

Solution

Substitution gives $\frac{0}{0}$.

Apply L'Hospital's rule:

$$\lim_{x \rightarrow 0} \frac{\cos x}{1} = 1$$

Example 2

Evaluate:

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$$

Solution

Indeterminate form: $\frac{0}{0}$

Differentiate numerator and denominator:

$$\lim_{x \rightarrow 0} \frac{e^x}{1} = 1$$

5. Solved Problems – $\frac{\infty}{\infty}$ Type

Example 3

Evaluate:

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{5x^2 - x}$$

Solution

Apply L'Hospital's rule:

$$\lim_{x \rightarrow \infty} \frac{6x + 2}{10x - 1}$$

Apply again:

$$\lim_{x \rightarrow \infty} \frac{6}{10} = \frac{3}{5}$$

6. Handling Other Indeterminate Forms

6.1 $0 \cdot \infty$ Type

Convert to fraction form.

Example

$$\lim_{x \rightarrow 0} x \ln x$$

Rewrite:

$$= \lim_{x \rightarrow 0} \frac{\ln x}{\frac{1}{x}}$$

Apply L'Hospital's rule.

6.2 $\infty - \infty$ Type

Combine terms into a single fraction.

Example

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x)$$

Multiply and divide by conjugate.

6.3 Exponential Forms

For:

- 0^0
- 1^∞
- ∞^0

Take **logarithm** and convert to product/quotient form.

7. Solved Problem (Exponential Type)

Example 4

Evaluate:

$$\lim_{x \rightarrow 0} (1+x)^{1/x}$$

Solution

Take logarithm:

$$y = (1+x)^{1/x} \Rightarrow \ln y = \frac{\ln(1+x)}{x}$$

Apply L'Hospital's rule:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1/(1+x)}{1} &= 1 \\ \Rightarrow \ln y &= 1 \Rightarrow y = e \end{aligned}$$

10. Practice Problems (Exam-Oriented)

1. $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$
2. $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$
3. $\lim_{x \rightarrow 0} x^x$
4. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$

PARTIAL DIFFERENTIATION & TOTAL DERIVATIVE

1. Introduction to Multivariable Functions

A function involving **more than one independent variable** is called a multivariable function.

Examples:

$$z = f(x, y), z = f(x, y, u), w = f(x, y, z)$$

Such functions are common in:

- Optimization problems
- Computer graphics
- Machine learning
- Image processing

2. Partial Differentiation

2.1 Definition

Partial differentiation is the differentiation of a multivariable function with respect to **one variable, keeping the other variables constant**.

If:

$$z = f(x, y)$$

then:

- Partial derivative w.r.t. x is denoted by $\frac{\partial z}{\partial x}$
- Partial derivative w.r.t. y is denoted by $\frac{\partial z}{\partial y}$

3. First Order Partial Derivatives

3.1 Definition

For a function $z = f(x, y)$:

$$\frac{\partial z}{\partial x} = \text{Derivative of } z \text{ treating } y \text{ as constant}$$

$$\frac{\partial z}{\partial y} = \text{Derivative of } z \text{ treating } x \text{ as constant}$$

Example 1

Let:

$$z = x^2y + 3xy^2$$

Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

Solution

$$\frac{\partial z}{\partial x} = 2xy + 3y^2$$

$$\frac{\partial z}{\partial y} = x^2 + 6xy$$

4. Second Order Partial Derivatives

4.1 Definition

Second order partial derivatives are obtained by differentiating the first order derivatives.

Notations:

$$\frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial y^2}, \frac{\partial^2 z}{\partial x \partial y}, \frac{\partial^2 z}{\partial y \partial x}$$

Example 2

For:

$$\begin{aligned}z &= x^2y + 3xy^2 \\ \frac{\partial^2 z}{\partial x^2} &= 2y \\ \frac{\partial^2 z}{\partial y^2} &= 6x \\ \frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial}{\partial y}(2xy + 3y^2) = 2x + 6y \\ \frac{\partial^2 z}{\partial y \partial x} &= \frac{\partial}{\partial x}(x^2 + 6xy) = 2x + 6y\end{aligned}$$

5. Equality of Mixed Partial Derivatives (Clairaut's Theorem)

If second order partial derivatives are **continuous**, then:

$$\boxed{\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}}$$

This is known as **Clairaut's theorem**.

6. Total Derivative

6.1 Concept

If a function depends on variables which themselves depend on another variable, then the derivative is called a **total derivative**.

6.2 Total Derivative of Two Variables

If:

$$z = f(x, y)$$

where:

$$x = x(t), y = y(t)$$

Then total derivative of z with respect to t is:

$$\boxed{\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}}$$

7. Total Differential

The total differential of $z = f(x, y)$ is:

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

This is useful in:

- Error estimation
- Approximation techniques

8. Solved Problems (Total Derivative)

Example 3

If:

$$z = x^2 + y^2$$

where:

$$x = \sin t, y = \cos t$$

Find $\frac{dz}{dt}$.

Solution

$$\begin{aligned}\frac{\partial z}{\partial x} &= 2x, \frac{\partial z}{\partial y} = 2y \\ \frac{dx}{dt} &= \cos t, \frac{dy}{dt} = -\sin t \\ \frac{dz}{dt} &= 2x \cos t + 2y(-\sin t)\end{aligned}$$

Substitute $x = \sin t, y = \cos t$:

$$\frac{dz}{dt} = 2 \sin t \cos t - 2 \cos t \sin t = 0$$

9. Composite Functions

A function involving **functions inside another function** is called a composite function.

Example:

$$z = f(u, v)$$

where:

$$u = g(x, y), v = h(x, y)$$

9.1 Chain Rule

The partial derivatives are:

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y}$$

10. Solved Example (Composite Function)

Example 4

Let:

$$z = u^2 + v^2$$

where:

$$u = x + y, v = x - y$$

Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

Solution

$$\frac{\partial z}{\partial u} = 2u, \frac{\partial z}{\partial v} = 2v$$

$$\frac{\partial u}{\partial x} = 1, \frac{\partial v}{\partial x} = 1$$

$$\frac{\partial z}{\partial x} = 2u + 2v = 2(x + y) + 2(x - y) = 4x$$

Similarly:

$$\frac{\partial z}{\partial y} = 2u(1) + 2v(-1) = 2(x + y) - 2(x - y) = 4y$$

13. Practice Problems

1. Find $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$ for

$$z = e^{xy}$$

2. If $z = x^2y + y^3x$, find total differential

3. Find $\frac{dz}{dt}$ if

$$z = xy, x = t^2, y = \sin t$$

JACOBIAN & MAXIMA-MINIMA OF FUNCTIONS OF TWO VARIABLES

1. Introduction

In multivariable calculus, it is often necessary to:

- Change variables from one set to another, and
- Locate **maximum and minimum values** of a function of two variables.

These tasks are carried out using:

- **Jacobian**
- **Maxima and Minima techniques**

Both topics are **high-weight VTU exam areas**.

2. Jacobian

2.1 Definition

If:

$$u = u(x, y), v = v(x, y)$$

Then the **Jacobian of (u, v) with respect to (x, y)** is defined as:

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

2.2 Expanded Form

$$\frac{\partial(u, v)}{\partial(x, y)} = \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x}$$

3. Interpretation of Jacobian

- Jacobian represents **rate of change** of variables
- Used for **change of variables**
- Helps in checking **functional dependence**
- Widely used in:
 - Image transformations
 - Coordinate transformations
 - Engineering simulations

4. Important Jacobian Properties

Property 1

$$\frac{\partial(x, y)}{\partial(x, y)} = 1$$

Property 2

$$\frac{\partial(u, v)}{\partial(x, y)} \cdot \frac{\partial(x, y)}{\partial(u, v)} = 1$$

(If inverse transformation exists)

5. Solved Problems – Jacobian

Example 1

If:

$$u = x^2 + y^2, v = 2xy$$

Find:

$$\frac{\partial(u, v)}{\partial(x, y)}$$

Solution

$$\frac{\partial u}{\partial x} = 2x, \frac{\partial u}{\partial y} = 2y$$

$$\frac{\partial v}{\partial x} = 2y, \frac{\partial v}{\partial y} = 2x$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 2x & 2y \\ 2y & 2x \end{vmatrix} = 4x^2 - 4y^2$$

Example 2 (Change of Variables)

If:

$$u = r \cos \theta, v = r \sin \theta$$

Find:

$$\frac{\partial(u, v)}{\partial(r, \theta)}$$

Solution

$$\begin{aligned}\frac{\partial u}{\partial r} &= \cos \theta, \frac{\partial u}{\partial \theta} = -r \sin \theta \\ \frac{\partial v}{\partial r} &= \sin \theta, \frac{\partial v}{\partial \theta} = r \cos \theta \\ \frac{\partial(u, v)}{\partial(r, \theta)} &= \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r\end{aligned}$$

6. Maxima and Minima of Functions of Two Variables

6.1 Definition

A function $f(x, y)$ has a **maximum or minimum** at a point if its value there is respectively:

- Greater than nearby values (maximum)
- Less than nearby values (minimum)

7. Necessary Conditions for Extrema

Let:

$$z = f(x, y)$$

First-order partial derivatives:

$$f_x = \frac{\partial z}{\partial x}, f_y = \frac{\partial z}{\partial y}$$

Necessary Condition

At an extremum point:

$$f_x = 0, f_y = 0$$

8. Second-Order Partial Derivatives

Define:

$$\begin{aligned}f_{xx} &= \frac{\partial^2 z}{\partial x^2} \\ f_{yy} &= \frac{\partial^2 z}{\partial y^2} \\ f_{xy} &= \frac{\partial^2 z}{\partial x \partial y}\end{aligned}$$

9. Second Derivative Test

Compute:

$$D = f_{xx}f_{yy} - (f_{xy})^2$$

Conditions

1. If $D > 0$ and $f_{xx} > 0$
→ Minimum value
2. If $D > 0$ and $f_{xx} < 0$
→ Maximum value
3. If $D < 0$
→ Saddle point
4. If $D = 0$
→ Test fails

10. Solved Problems – Maxima & Minima

Example 3

Find the extrema of:

$$z = x^2 + y^2 + 2x + 4y$$

Solution

$$\begin{aligned}f_x &= 2x + 2 = 0 \Rightarrow x = -1 \\f_y &= 2y + 4 = 0 \Rightarrow y = -2\end{aligned}$$

Second derivatives:

$$\begin{aligned}f_{xx} &= 2, f_{yy} = 2, f_{xy} = 0 \\D &= (2)(2) - 0 = 4 > 0\end{aligned}$$

Since $f_{xx} > 0$, function has a **minimum** at $(-1, -2)$.

Minimum Value

$$z_{\min} = (-1)^2 + (-2)^2 - 2 - 8 = -5$$

11. Example 4 (Saddle Point)

Find nature of:

$$z = x^2 - y^2$$

Solution

$$f_x = 2x, f_y = -2y$$

At $(0, 0)$:

$$\begin{aligned}f_{xx} &= 2, f_{yy} = -2, f_{xy} = 0 \\D &= -4 < 0\end{aligned}$$

Hence, **saddle point** at origin.

14. Practice Problems

1. Find extrema of $z = x^2 + y^2 - 4x - 6y$

2. Find nature of $z = x^3 - 3xy^2$

3. Evaluate Jacobian for $u = xy, v$

$$= x/y$$